

Solar Water Heating Up

A wide range of options exist for ready-to-install, readily certified solar water heating systems. To further heat things up, government and utility incentives are expanding.

by William T. Guiney

Solar water heating is once again gaining market share, and with good reason. The development of HERS and Energy Star, together with technological refinements, is transforming the way home builders look at water heating options. Complete off-the-shelf systems have evolved and are being readily certified for a myriad of rebate programs.

In most applications, solar water heating contributes more to saving energy than any other solar technology. Solar water heating at \$1.00–\$2.50/W is a better investment than photovoltaics at \$6.00–\$10.00/W. Solar domestic hot water (SDHW) systems in the market cost \$1,000–\$2,500/kW for a typical residential system.

According to the national Solar Rating and Certification Corporation, a typical residential SDHW system for a family of four delivers 4 kW of electrical equivalent thermal power under full sun, and when the temperature of the water in the storage tank is about the same as the air temperature. Such a system typically has about 64 ft² of solar collector surface area and produces approximately the same peak power as 400 ft² of PV panels.

Because peak performance occurs infrequently, a more realistic indication of solar thermal system performance is the rated daily energy output of the collectors or system. Using this method, a typical SDHW system contributes 7–10 kWh per day, depending on the solar resource and the type of collector or system.

SDHW is the better energy investment if the backup water heating is electric-resistance heat; if one uses a



HELIODYNE, INC.

heat pump or gas-based backup, the economics may vary.

Many new system sales today are to repeat customers who have moved and want new systems installed on their new homes. After 15–20 years of experience with solar water heating systems, these homeowners know the value and performance that these systems deliver. And the newer systems deliver even more value than the older ones as, since the 1980s, the solar water heating industry has seen the development of better systems, certified installers, lower prices, and proven performance. The familiarity with solar water heating, anticipated savings, and confidence in the systems' performance have led a number of state weatherization programs to incorporate solar as part of their low-income energy savings

measures (see "Making Solar Hot Water Affordable," *HE* Jan/Feb '99, p. 8).

Options for SDHW

Although the array of system choices may seem confusing at first, the choice of a solar water heating system is quite simple. The most important factor is the climate where the system is to be installed. But before I elaborate on climate types, I need to explain four basic solar system terms: direct, indirect, passive, and active.

Direct solar systems heat the water that is consumed or stored in the water heater (see Figures 1, 2, and 3). If the water in the storage tank or water heater is directly heated in the solar collector, it is a direct system. In other

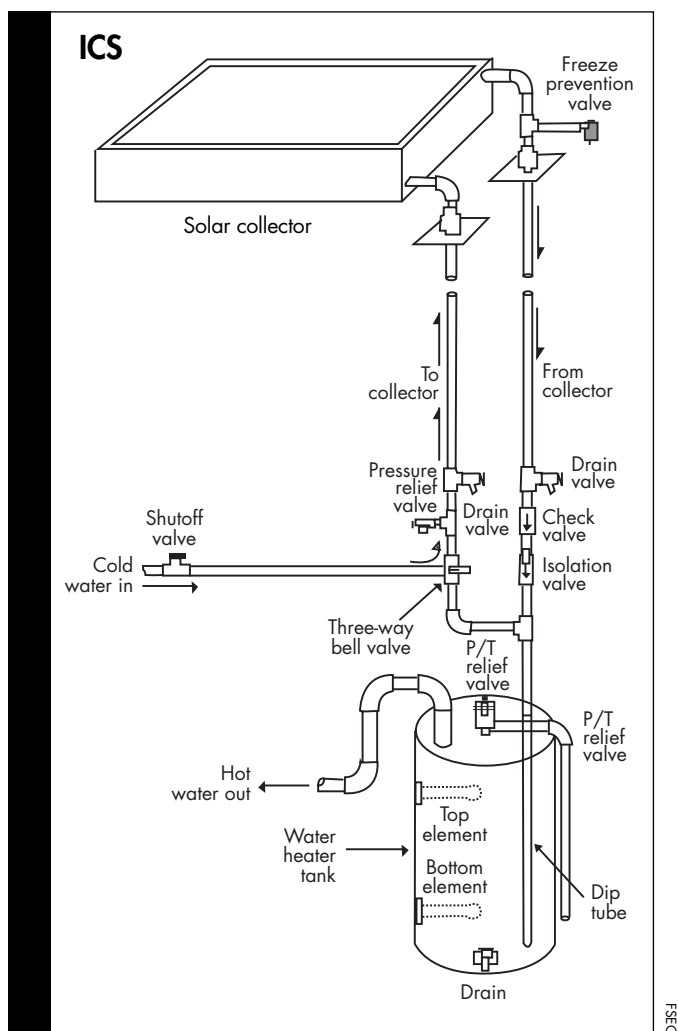


Figure 1. No pumps or controllers are necessary for an integral collector storage system (ICS).

words, the water you use in the house is the same water that has been circulated through the collectors.

Indirect systems incorporate a heat exchanger separating the solar collector from the storage tank. An antifreeze solution circulates through the collector, and the heat exchanger transfers the heat from the antifreeze solution to the potable water in the tank. This heat exchanger can be internal or external to the storage tank.

If a solar water heating system has a circulation pump to transfer heat from the collector to the storage tank, the system is classified as an active system. If the system has no pump or control mechanism to transfer the heat to the storage tank, it is a passive system.

on the roof, so the solar water heating industry has developed systems with storage tanks that are recessed into the attic or located on areas of the roof not visible from the front of the house, thus providing a low profile for the system (see Figure 2).

The simplest solar water heater is the integral (or integrated) collector storage system (ICS), also referred to as the batch or bulk (or breadbox) storage system (see Figure 1). In these systems, the collector and the storage are incorporated into one unit. The hot water generated by ICS systems should be used during the afternoon and evening, so that the stored hot water is consumed or transferred into a separate storage tank in the house before dark.

Passive systems use no pumps. Instead, they use natural forces like thermosiphon. In thermosiphon systems, thermosiphon action occurs when water heated in the collector becomes more buoyant than the colder water from the storage tank. The heavier, descending, colder water from the tank forces the heated water back to the tank. This is the most popular system design used outside the United States. Travel anywhere in the Caribbean, the Middle East, the Mediterranean, Australia, or Asia, and you will see the thermosiphon system. In these areas, solar has long been considered the only option for heating water in homes and small businesses. Many Americans dislike seeing the thermosiphon tank

This will reduce overnight heat loss during colder weather. The water is used or transferred when someone makes a draw in the house. When a draw is taken, the water in the ICS unit replaces the water that was used from the backup tank below.

Climate Matters

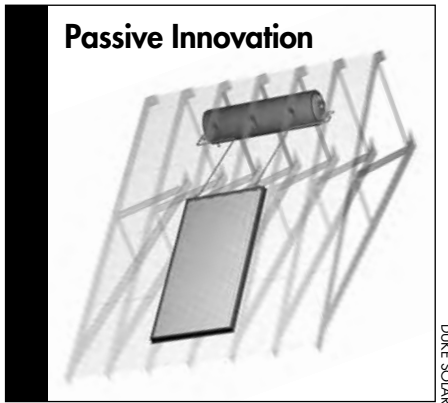
Direct systems should be limited to warm climates or to areas that experience only a couple of freezing days per year. And those freezing days should include only a few hours' worth of below-freezing temperatures. Direct systems are only intended to be installed in areas that have occasional freezes in the winter, not sustained days or long, clear nights below 40°F.

If you live anywhere that experiences regular seasonal freezing weather (most of the United States, except for parts of Florida and California), you need an indirect system. Indirect solar water heating systems incorporate a heat exchanger to transfer the heat energy from the collector to the potable water in the water heater. Freeze proof heat transfer fluids are used in the indirect system collector loop to protect the collector during freeze conditions. Not all indirect systems use antifreeze solutions; some systems have a drain-back design. This design strategy allows all of the heat exchange fluid to drain out of the solar collector into a reservoir when the pump stops circulating. Therefore, there is no solution in the solar collector to be damaged by freezing conditions.

Backing Up the System

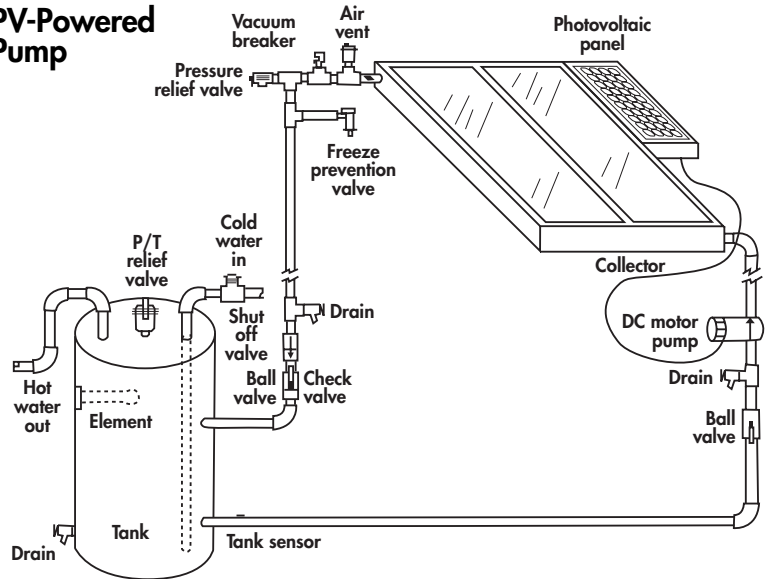
What about days when it's overcast or there isn't enough sun to heat the water to the desired use temperature? What type of backup water heater should you choose? This is a very important decision (see "Solar Water Heating," *HE* Jan/Feb '02, p. 6). What are the typical water heating options in the area—electricity, natural gas, LPG or propane? All of these can be used as the back-up heating fuel. Most solar storage tanks have an electric element that heats only a portion of the storage tank when necessary, to increase

Passive Innovation



(above) **Figure 2.** This new passive freeze proof design system includes a roof-integrated collector and 316-l storage tank. (right) **Figure 3.** In a direct system with a PV-powered pump, water will flow through the collector only when the sun is shining.

PV-Powered Pump



the potable water to the desired use temperature. A demand water heater will reduce electric water heater tank losses and will save space in the house.

If gas is used in anything other than a instantaneous-demand heater, you should choose a two-tank system. This system consists of a standard gas water heater and a solar storage tank. Because a typical gas water heater has the burner located at the bottom of the tank, whenever hot water is used in the house, cold water entering the tank turns the burner on, which in turn heats the whole tank. To use a solar system with gas as a backup, you must have a separate solar storage tank that has the cold service water entering the system. The solar storage tank will supply preheated water to the typical gas backup heater. Thus the gas burner will run only when it is needed to boost the temperature of the preheated water for delivery into the house. Another advantage of this system is that during the summer the gas heater can be bypassed using valve strategies, eliminating its high standby heat loss.

Many active solar water heating systems use a PV module to generate electricity to run a DC pump motor (see Figure 3). This is one of the better uses for PV; the current generated by the

PV panel controls the speed of the pump. This has a direct relationship to the amount of heat energy being generated by the solar thermal collector. This combination is quite elegant: no controller, no battery, no inverter, no grid connection! Used in a solar water heating system, water pumping becomes one of the most cost-effective PV applications.

Government and Utility Support

The development of HERS, the Energy Star programs, and many state energy programs (including those in Florida, Illinois, Arizona, California, and North Carolina, where there is an attractive 35% tax credit for solar water heating) is helping to highlight the efficacy of solar DHW. While I was at the Florida Solar Energy Center, I taught and promoted the Florida Energy Gauge (HERS) program. I demonstrated the advantages of including a SDHW system in a home to qualify for the Energy Star Home program using the Engauge ratings software. In many cases, a simple radiant barrier, a 12-SEER heat pump, and a SDHW system qualified a typical production-built home for the Energy Star label.

Many utility programs are offering rebates for solar water heating. There are also potential federal tax credits for solar water heating.

Home builders are now learning the advantages of incorporating solar water heating systems into new homes. The renewable portfolio standards in some states will allow solar water heating to demonstrate the reliability and performance of today's systems and offer real savings to buyers and lower risks for contractors and financiers.

Consumers have shown that, given the opportunity, they will choose a solar water heating system. For new construction, working with a solar contractor to identify subdivision lots with good orientation and models that are aesthetically pleasing with new skylight-styled solar water heating systems can simplify the sales process and narrow the options for the builder and consumer. Combining tax credits for the builders of energy-efficient homes and credits for the purchase of solar water heating systems could increase demand nationally.

New Challenges

I would imagine that a few firms have a couple of new designs waiting for the right market opportunities. A solar water heating system may cost

\$2,000–\$4,000 for the equipment, plus the cost of installation, today— but maybe that long-awaited \$1,000–\$1,500 system is out there on the drawing boards of more than a couple of engineers. Maybe someone is even building it right now.



William T. Guiney is the Water Heating Division manager of Duke Solar Energy, LLC, in Raleigh, North Carolina. Prior to joining Duke Solar in 2001, Bill was a solar thermal specialist and instructor for the Florida Solar Energy Center.

For more information:

To find certified installers across the country, go to www.seia.org.

The Database of State Incentives for Renewable Energy (DSIRE, at www.dsireusa.org) provides a wealth of information on incentives, programs, proposed policies, and existing legislation that promote renewable energy. DSIRE, a project of the Interstate Renewable Energy Council (IREC), is funded by DOE's Office of Power Technologies and managed by the North Carolina Solar Center.

Annual, site-specific energy savings for domestic water heating systems are available at www.solar-rating.org for all systems certified by the Solar Rating and Certification Corporation. According to these data, a typical SDHW produces about 3,400 kWh per year, depending on local conditions and type of collector. Additional useful information may be found at the FSEC Web site (www.fsec.ucf.edu); and the Florida SWAP program Web site (www.alpha.fsec.ucf.edu/swap/).

Two of the manufacturers in this field are heliodyne (www.heliodyne.com) and Duke Solar (www.dukesolar.com).